Divergence

2

$$\nabla \cdot \vec{v} = \left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}\right)$$

Analogous to

Fluids → $\nabla \cdot \vec{v} = 0$

if fluid in = fluid out in a particular volume. ⟹

incompressible fluid (Analogy)

incompressible phase flow (velocity field here)

Minus First Law (Susskind's)

i.e. satisfies:

Derive Hamilton's Eq.^{n}

$$\mathcal{H} = \sum_i (p_i \dot{q}_i) - \mathcal{L}(q_i, \dot{q}_i)$$

$$\delta \mathcal{H} = \sum_i \left(\dot{q}_i \delta p_i + p_i \delta \dot{q}_i - \frac{\partial \mathcal{L}}{\partial q_i}\delta q_i - \frac{\partial \mathcal{L}}{\partial \dot{q}_i}\delta \dot{q}_i\right)$$

$$\delta \mathcal{H} = \sum_i \left(\dot{q}_i \delta p_i - \frac{\partial \mathcal{L}}{\partial q_i}\delta q_i\right) \quad -(i)$$

($\frac{\partial \mathcal{L}}{\partial q_i} \to \dot{p}_i$)

$$\delta \mathcal{H}(q_i, p_i) = \sum_i \left(\frac{\partial \mathcal{H}}{\partial p_i}\delta p_i + \frac{\partial \mathcal{H}}{\partial q_i}\delta q_i\right) \quad -(ii)$$

Comparing (i) ; (ii) :

$$\frac{\partial \mathcal{H}}{\partial p_i} = \dot{q}_i \;;\; \frac{\partial \mathcal{H}}{\partial q_i} = -\dot{p}_i$$

This incompressibility of the phase space fluid is called:

Liouville's Theorem

no. of coord = N ⟹ 2N dimensional phase space $(p_i ; q_i)$

⟹ 2N components of $\vec{v}$ field (v_{p_i}, v_{q_i}) → $(\dot{p}_i, \dot{q}_i)$ → $\left(-\frac{\partial \mathcal{H}}{\partial q_i}, \frac{\partial \mathcal{H}}{\partial p_i}\right)$ (from H's eq.^{n})

$$\nabla \cdot \vec{v} = \sum_i \left(\frac{\partial v_{q_i}}{\partial q_i} + \frac{\partial v_{p_i}}{\partial p_i}\right) = 0$$ (if fluid is incompressible)

$$\nabla \cdot \vec{v} = \sum_i \left(\frac{\partial}{\partial q_i}\frac{\partial \mathcal{H}}{\partial p_i} + \frac{\partial}{\partial p_i}\frac{\partial \mathcal{H}}{\partial q_i}\right)$$

Derive H

1

$\mathcal{L}(q_i, \dot{q}_i, t)$ ← explicit

$$\frac{d\mathcal{L}}{dt} = \left(\frac{\partial \mathcal{L}}{\partial q_i}\dot{q}_i + \frac{\partial \mathcal{L}}{\partial \dot{q}_i}\ddot{q}_i\right) + \frac{\partial \mathcal{L}}{\partial t}$$ → 0 (if implicit)

$$= \dot{p}_i \dot{q}_i + p_i \ddot{q}_i$$

$$\frac{d\mathcal{L}}{dt} = \frac{d}{dt}(p_i \dot{q}_i)$$

$$\frac{d}{dt}(\mathcal{L} - p_i \dot{q}_i) = 0$$

$$\boxed{\mathcal{L} - p_i \dot{q}_i = \text{const}} = \mathcal{H}$$

$$\frac{1}{2}m\dot{q}_i^2 - V(q_i) - m\dot{q}_i^2$$

$$-\left(\frac{1}{2}m\dot{q}_i^2 + V(q_i)\right) = \text{const.}$$

aka. E_s = const

$$\mathcal{H} = \sum_i \dot{q}_i \frac{\partial \mathcal{L}}{\partial \dot{q}_i} - \mathcal{L}$$

Example - 1.1

→ $\dot{q} = \sqrt[4]{km}\,\dot{x}$

⟹ $\dot{q}^4 = km\dot{x}^4$

⟹ $\dot{x}^2 = \frac{\dot{q}^2}{\sqrt{km}}$

$\omega = \sqrt{\frac{k}{m}}$

$$\mathcal{L} = \frac{m\dot{x}^2}{2} - \frac{kx^2}{2}$$

$$= \frac{\dot{q}^2}{2\omega} - \frac{\omega}{2}q^2$$

$$\boxed{\mathcal{L} = \frac{\dot{q}^2}{2\omega} - \frac{\omega q^2}{2}}$$

Example - 1.2 convert to H

$$\mathcal{L} = \frac{\dot{q}^2}{2\omega} - \frac{\omega}{2}q^2$$

$$\mathcal{H} = p\dot{q} - \mathcal{L}$$

$$= p\dot{q} - \frac{\dot{q}^2}{2\omega} + \frac{\omega}{2}q^2$$

$$\mathcal{H} = \frac{\dot{q}^2}{\omega} - \frac{\dot{q}^2}{2\omega} + \frac{\omega}{2}q^2$$

$$= \frac{\omega \dot{q}^2}{2\omega^2} + \frac{\omega}{2}q^2$$

$$\mathcal{H} = \frac{p^2\omega}{2} + \frac{q^2\omega}{2} \quad \left[\because p = \frac{\dot{q}}{\omega}\right]$$

$$\mathcal{H} = \frac{\omega}{2}(p^2 + q^2)$$

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